

ASHWANI YADAV

B.S. - M.S., Mathematics & Computing, IIT Kanpur '19

+91-7860-206-546
ashwani.yadav1952@gmail.com
[github:ashwani-09](#)
[linkedin:ashwani1952](#)

WORK EXPERIENCE

HSBC AM/ DATA ANALYTICS

JUL 2019 - PRESENT

Environmental-Social-Governance (ESG) Framework

- Built the first-ever centralized ESG framework end-to-end to calculate ESG scores of **330K equity & FI** financial instruments
- Developed PySpark based engine for efficient sourcing and aggregation of large data on multi-node **GCP Dataproc** cluster
- Trained and fine-tuned a BERT based financial text **sentiment model** to identify potential future ESG score changes
- Designed **interactive dashboards** to view and compare company/portfolio ESG metrics vs parent benchmarks & competitors
- Improved framework run-time by **40%** utilising **BigQuery engine & Spark configs** in Dataproc clusters

Data Quality (DQ) Framework

- Designed and developed an advanced analysis framework to calculate the data quality measures of internal data sources across industry standard DQ dimensions such as **Completeness, Conformity, Validity, Timeliness & Uniqueness**
- Integrated the framework for 4 critical internal data sources covering **400 CDEs** in more than **900 dimensional rules**
- Designed interactive **executive summary dashboards** on Tableau server for data owners to view up-to-date DQ measures
- Prepared the **code documentation** of entire project in Github covering all module definitions and dependencies

Venture Capital Investments

- Performed **API testing** using Python to test functionality, performance & reliability of vendor data for the VC fund team
- Researched competition & investment landscape of early-stage startups using **network & exploratory analysis**
- Developed a statistical **machine learning model** (PoC) which predicts the likelihood of success of early-stage startups
- Developed data-ingestion (ETL) pipelines to onboard vendor data into **GCP BigQuery** tables and **Cloud Storage** buckets

SHAREKHAN / DATA SCIENCE INTERN

MAY 2018 - JUL 2018

Equity-to-First SIP: Cross Sell Model

- Built a **ML ensemble model** in R which predicts the propensity of existing equity clients to start investing in mutual funds
- Deployed a list of high propensity clients & best suited communication channels to sales teams for **A/B testing** of model
- Researched the SIP clients's behaviours and created a generic **SIP client profile** through statistical exploratory analysis
- Derived the investor sentiment indicators & relevant demographic features of **770K clients** from **MS-SQL servers**

TECHNICAL SKILLS

- | | | |
|-------------------------------|-----------|-----------|
| - Python | - Git | - Tableau |
| - Google Cloud Platform (GCP) | - PySpark | - MS-SQL |

Courses & Certifications

- Achieving Advanced Insights with BigQuery by Google Cloud on Coursera [[Certificate Link](#)]
- Applying Machine Learning to your Data with GCP by Google Cloud on Coursera [[Certificate Link](#)]

Virtual MF Portfolio Analytics Tool (Personal Project)

- Building a portfolio analytics tool for individual investors to **monitor performance** and **identify risks & growth opportunities**
- Developing features to **identify correlation of portfolio investments** in current portfolio as well as in virtually added funds
- Built a **data scraping** pipeline to extract MF holdings data from a public website in Python with ``requests`` & ``beautifulsoup``

EXTRACURRICULARS.....

Initiatives at HSBC

- **Monthly presentations** with team members to share new learnings on Python ecosystem, ML models, GCP & Tableau
- Workshops and clinics for **business teams** to help them adapt to GCP ecosystem & Standard SQL and follow best practices

TECHKRITI, IIT KANPUR

MAY 2016 - APR 2017

Head, Marketing Team

- Lead a 3-tier marketing team of 50 executives for sponsorship & promotion of IIT Kanpur's annual technical & entrepreneurial festival
- Coined & organised H.O.P.E., an educational initiative for underprivileged students, in collaboration with Jagran iNext & local NGOs
- Revived and promoted international culture & education fair in Techkriti by actively collaborating & meeting with 4 country embassies
- Secured biggest title sponsorship and largest overall cash sponsorship budget, INR 10Mn, among all technical festivals in North India
- Expanded Technocruise (technical workshops) & TOSC (school level science competition) to 11 new cities in addition to last year's 4

COUNSELLING SERVICE, IIT KANPUR

MAY 2015 - APR 2016

Student Guide

- Supervised and organised annual fresher's orientation programme for 800+ students and helped them blend into the campus culture
- Guided 6 counselees throughout the year in acclimatising to the campus realm & counselled them during mental & academic worries